ANSI/IES LM-48-20

Illuminating
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# APPROVED METHOD: TESTING THE CALIBRATION OF LOCKING-TYPE PHOTOELECTRIC CONTROL DEVICES USED IN OUTDOOR APPLICATIONS

## AN AMERICAN NATIONAL STANDARD

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# APPROVED METHOD: TESTING THE CALIBRATION OF LOCKING-TYPE PHOTOELECTRIC CONTROL DEVICES USED IN OUTDOOR APPLICATIONS

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has been approved by IES.
Suggestions for revisions
should be directed to IES.

**Prepared by**
**The IES Testing Procedures Committee**

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# Forward

This Lighting Measurement (LM) document is a revision of *IES LM-48-01, IES Guide for Calibration of Photoelectric Control Devices*. In addition to the guidelines for testing the response of photoelectric control devices, supplemental information is provided for the construction of a test apparatus suitable for such testing.

# 1.0 Introduction and Scope

## 1.1 Introduction

Photoelectric control devices, also called *photo controls*, are light-sensing devices that, in response to changing ambient light levels, either close or open a load-carrying switch to control the flow of electric current to a load, usually a lighting device. Calibration under real-world outdoor conditions (i.e., in a roof-top facility under actual daylight and the prevailing weather) entails expensive, time consuming procedures not practical for commercial use. Actual ambient light conditions at levels where photo controls normally switch state can vary widely in terms of correlated color temperature and spectral power distribution. To duplicate these various conditions in a test setup is unrealistic and impractical. A "best compromise" design has led to the present Approved Method using a tungsten-halogen lamp in combination with an infrared blocking filter and a color correction filter.

## 1.2 Scope

The objective of this Approved Method is to describe a procedure and test equipment by which photoelectric control devices can be tested to obtain accurate, optimally comparable data. The system aspects that need to be understood to calibrate light sensitive control devices used in roadway and outdoor area lighting are covered. Although photo controls are often used in indoor applications, the method of testing the calibration described in this Approved Method is limited to photo controls used in outdoor applications.

No attempt has been made to suggest turn-on and turn-off levels, nor to develop test procedures for the mechanical and electrical components. These items are fully covered in *ANSI C136.10-2017, Locking-type Photo control Devices and Mating Receptacle Physical and Electrical Interchangeability and Testing*, which should be consulted for a comprehensive investigation of the selection, calibration, and testing of photoelectric control devices.[1] The test procedures described in this Approved Method are limited to testing the photometric calibration of the device only.

# 2.0 Operating Conditions

The following variables need to be considered when building an instrument for testing the calibration of photoelectric control devices:

- *Color*. Photo controls operate over a wide range of spectral power distributions between a clear day and a rainy day, between sunset and dark, and between sunrise and daytime. Time of year, location, and control orientation also cause changes in the sky spectral energy reaching the photo detector.
- *Rate of change of illuminance levels*. Illuminance levels vary rapidly at sunset and sunrise, covering a range of between 0 and 300 lux (approximately 0 to 30 fc) in approximately 30 minutes.
- *Field of view*. Many designs of photo controls exist, each with its own peculiar optical arrangement for light collection. The method of adjustment for setting individual controls imposes further variations on the optical arrangements. Most photo controls have sensors looking at the horizon; however, designs exist where the photocell faces upward. An appropriate factor to convert north-sky vertical illuminance to horizontal illuminance has been found to be approximately 1.4 (i.e., $E_h$ is equal to $1.4E_v$).[2]
- *Temperature*. Photo controls are designed to operate over a wide range of outdoor temperatures (-40 °C to 65 °C [-40 °F to 149 °F]).
- *Voltages*. It is critical that proper operation be attained over a wide range of line voltages. Photo controls are designed to operate either from a single line voltage or over a nominal voltage range from 120 to 480 volts. Line voltage variations of ±5 percent or more are likely to occur.

- *Cell types*. The light-sensitive detectors used in photo controls are usually of the uncorrected cadmium sulfide (CdS) type or the silicon (Si) type. CdS cells have a spectral response quite close to photopic vision (peak sensitivity between 520 and 560 nm), while Si cells have a response that peaks at around 850 nm (in the infrared IR-A range), considerably beyond the spectral range of human vision. Some Si photocells are equipped with an integral IR blocking filter, such as Schott KG-1 glass, to reduce the effects of changes in ambient morning or evening correlated color temperatures due to clouds, haze, or smoke, which can cause response drift and affect the turn-on and turn-off levels of the photo control. Some of the Si-based photo control designs omit the IR-A filter because of cost considerations.
- *Relay types*. Although not a subject of this Approved Method, it is important to point out that load relays used in photo controls are of two types, magnetic and thermal, which have considerably different characteristics. The *magnetic* load relay reacts relatively quickly to the current provided by the detector element and is basically insensitive to ambient temperatures. Three categories of electrical circuits are used with magnetic load relays, which switch the load instantaneously (delay less than 0.5 second), with a short delay (0.5 to 5 seconds), or with a delay of over 5 seconds. The *thermal* load relay uses a bimetal strip that, when heated by the current provided by the detector element, curves sufficiently to operate snap-action contacts. The inherent time required to heat and cool down is affected by ambient temperature conditions and might range from 15 to 120 seconds; therefore, it needs to be considered in setting the turn-on level of the photo control.

# 3.0 Test Equipment

## 3.1 General Objective

The test equipment should:

- Be easy to operate with only a minimum of operator instruction and be as free as possible from operator bias
- Require only a moderate space and be easy to maintain and relatively low in cost to build
- Be capable of testing the calibration to national standards for measurements

## 3.2 Accuracy

The test equipment should be capable of testing the calibration of the controls under test, such that the turn-on level of a north-facing control mounted at 10.7 meters (35 feet) on a clear, cloudless day may be verified to within ±10 percent.

## 3.3 Varying Levels

The test equipment should include a method of adjusting the light level to simulate sunset and sunrise, which ranges from 3 lux (approximately 0.3 fc) or less to 150 lux (approximately 15 fc) or higher.

## 3.4 Field of View

The test equipment should provide a surrounding condition for the control that is typical of situations of its use and such that the calibration verification will be adequate. The spherical acceptance angle of the photometer shall be equal to or exceed that of the photosensitive element of the photo control.

## 3.5 Operating Indication

The test equipment should have an indicator to show the switching on or switching off of the load contacts of the controls. The light level should remain constant at this point so that an ambient light level reading can be obtained, unless instrumentation is used to automatically record the light levels at the turn-on and turn-off points.

## 3.6 Color of Light Source

Since it is not practical to replicate the spectral content of daylight at dawn and dusk, the light source suggested for this application shall be a tungsten halogen lamp. The lamp shall be operated from a regulated DC power supply (±1.0%) and used with an IR-A blocking filter, such as Schott KG-1 glass filter or equivalent. In addition, a suitable color correction filter, such as a Hoya Optics LB-40, should be used to bring the correlated color temperature close to 4000 K.

## 3.7 Measuring Operating Level

The illuminance meter used to read operating levels shall be cosine corrected. It shall be color corrected to ±2.0% (CIE $f_1'$).[3] Operating levels are defined in lux as determined by measurements using the illuminance meter in a plane parallel to the plane of the photosensitive element of the photoelectric control under test. The illuminance meter shall have less than 5.0% error. It shall be capable of reading low light levels of 2 lux accurately. If desired, a calibrated photometer arrangement can be substituted in lieu of a readily available commercial illuminance meter.

### 3.8 Line Voltage Adjustment

The test equipment shall be adjustable to voltages over which the photo control is designed to operate. The alternating current (AC) voltage to the photo control shall be regulated such that the root-mean-square (RMS) of the harmonic components does not exceed 3.0% of the value of the fundamental frequency of the line voltage. It shall be held to ±2.0% of the voltage at which the photocontrol is being tested.

### 3.9 Load Current

Calibration shall be done with no load current, or performed with a load current equal to or less than 1 ampere. With a control device using a bimetal switch to carry the full load current, means should be provided to apply the full load current, as this could affect the turn-off level of the control.

### 3.10 Ambient Temperature

When testing photocontrols that are temperature sensitive, it is suggested to provide for variable ambient temperature control of the test chamber in which the photocontrol under test is located. Caution must be exercised in such cases to maintain the accuracy of the illuminance meter. Since it may take as much as 30 minutes for such controls to reach stable temperatures at the limits of the allowable temperature range, it is suggested to consider a separate pre-conditioning setup.

## 4.0 Test Procedures

### 4.1 Calibration

The illuminance meter shall be calibrated annually using standards of illuminance traceable to a national standards laboratory (such as NIST).

### 4.2 Cell Conditioning

Photo controls using CdS cells should be conditioned under a high light level (1,000 lux or more) for four hours prior to testing to remove all dark memory. Silicon cells do not have dark memory and require no cell conditioning.

### 4.3 Position of Device

The photo control and the illuminance meter shall be positioned in the calibrator so that each photosensitive element faces the light source of the calibrator. Using properly oriented locking type NEMA receptacles will ensure proper orientation of horizontally facing photo controls.[1]

### 4.4 Load

The testing procedure contained in this Approved Method is based on a "no-load" condition, except for bimetal switch types (thermal relays), which shall be tested with the full rated load current. Under actual field operating conditions with various loads (types of lamps and wattage), the turn-off might be affected.

### 4.5 Energize

The power to the photo control is turned on. The voltage applied shall be that for which the photo control is rated. For “single voltage" rated controls having a “±” range, the center (or nominal value) of that voltage range shall be used for the test. In case of a "multiple voltage" rated photo control, the test shall be performed twice: once, using the lowest voltage for which the photo controls is rated, and again, using the high end of the photo control's voltage rating.

### 4.6 Determining Turn-On and Turn-Off Levels

The light level is decreased slowly by mechanical means without changing the correlated color temperature to determine the turn-on setting for the control device or devices. The dynamic light level range should exceed the operational range of the photo control under test; i.e., a minimum upper level of 150 lux or higher for most photo controls, and a lower level as close to zero lux as practical. To determine the turn-off level, the light level is increased slowly until the device turns off. For multiple-voltage rated photo controls, this procedure is repeated for the minimum and maximum rated input voltages of the photo control.

The standard test ambient temperature should be 25 °C ± 5 °C (77 °F ± 9 °F). Tests may be run at other ambient temperatures and so reported. Controls of the "instant acting" type (no delay; response time less than 0.5 second) should be tested at a ramp rate of 1 lux per second. "Fast acting" controls (short delay; response time 0.5 to 5 seconds) should be tested at a ramp rate of 0.4 lux per second or less. "Slow acting" types (long delay; response time more than 5 seconds), should be tested at a ramp rate not faster than 0.1 lux per second. The ideal ramp speed for these controls would be found by using a somewhat compressed light-level decrease (or increase) simulating average natural lighting condition changes at dawn and dusk. These changes can be expressed as a third-order polynomial function:

$$Y_v \text{ (in lux)} = 0.1517x^3 - 0.1509x^2 + 1.9247x + 3.5339$$

or

$$Y_v \text{ (in fc)} = 0.0141x^3 - 0.0140x^2 + 0.1789x + 0.3284 ,$$

where $y$ represents the vertical north-sky illuminance (in lux or fc) and $x$ is time expressed in minutes (see **Figure 4-1**).

The formula compresses the time constant by a factor of two. This is suggested to keep the calibration procedure within a reasonable time limit. Unless the controls are of the inverse-ratio type (the turn-on level is higher than the turn-off level), there is no need to ramp the light above the turn-off level or below the turn-on level. Suggested turn-on and turn-off values for photo control calibrations can be found in ANSI C136.10-2017.[1]

## 4.7 Temperature Range

Since photo controls will be operated over a wide range of ambient temperatures, it might be necessary to perform some testing in an environmental test chamber capable of providing temperatures from -40 °C to 65 °C (-40 °F to 149 °F). For such tests, special attention needs to be paid to the temperature sensitivity of the illuminance meter used to monitor the light level inside the test box. For special applications, where the photo controls operate under extreme temperature conditions, an optional temperature-dependent calibration procedure shall be considered. Although photoelectric controls may operate and be tested at extreme temperatures, the illuminance meter used during the test needs to be

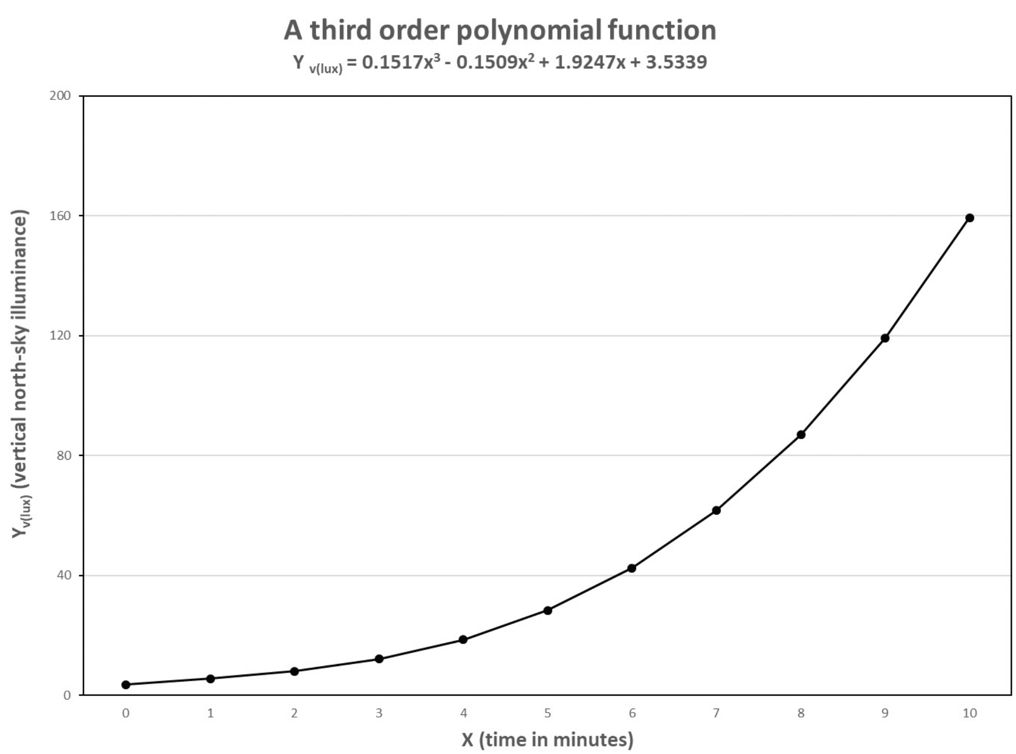



**Figure 4-1. Change in vertical north-sky illuminance level ($Y_v$, in lux) as a function of time ($x$).**
(© Illuminating Engineering Society)

either suitable for operating properly at such extremes or shall be operated at an ambient temperature for which it is designed.

## 5.0 Test Reports

### 5.1 Control Description

The control description shall include:

- Manufacturer's name
- Catalog number and/or adequate description for identification
- Voltage, current, wattage, and/or VA rating
- Turn-on light level
- Turn-off light level
- Ratio of turn-off light level to turn-on light level
- Test operating voltage
- Test load current
- Test ambient temperature
- Report of turn-on and turn-off changes with changes in photocontrol voltage from minimum to maximum rated voltages
- Date of test
- Laboratory performing the test

*Note:* The test report should contain any information about the test setup and execution that would aid in the exact reproduction of the test at a future date.

## Annex A – Suggested Apparatus for Testing the Calibration of Locking-Type Photoelectric Control Devices Used in Outdoor Applications

### A.1 General

A "test box" was designed and constructed to verify the method described in this Approved Method. Such a test box allows checking the turn-on and turnoff settings of photoelectric control devices. The system utilizes uniform illuminance towards the photoelectric control and an integral illuminance meter with a remote read-out. Spatial uniformity of the source is ensured by the use of two sheets of opal glass at optimal spacing, along with a unique shutter assembly. This allows the correlated color temperature (CCT) of the light source to stay constant during the test. By connecting a recorder to the analog output of the integral illuminance meter, turn-on and turn-off values can be recorded.

The proposed apparatus is relatively slow, taking about two minutes for the testing of one photoelectric control device of the "instant acting" or "fast acting" type. For "slow acting" controls using a suggested stepper-motor drive and control, one complete test cycle is expected to take as long as 25 minutes. (Such a design has not been fully investigated or tested.)

### A.2 Design and Exceptions

To ensure that the procedure spelled out in the Approved Method is sound and can be performed with reasonable accuracy, a prototype test apparatus was constructed to verify the procedure. This test apparatus is depicted in **Figures A-1** through **Figure A-5** (see **Section A.3**). The first four figures include building materials lists. Since this prototype was built only to ensure that the concept of this Approved Method was sound, many of the parts used were items on hand. The IES neither suggests nor endorses the use of any particular manufacturer's product; therefore, any listed items that are obtainable from other sources or manufacturers may be used. The 0.64-cm (¼-in.) thick aluminum plate used to construct the test box was on hand when the test apparatus was built. Such heavy stock is not necessary; using a thinner plate will reduce both the weight and cost of the apparatus. The friction drive of the shutter wheel can also be replaced by a positively engaging mechanism (gear-drive type).

The suggested shutter wheel is quite suitable for testing "instant acting" and "fast acting" photo controls. To change the speed at which the light level increases or decreases (necessary to check "fast acting" photo controls), the rotational speed of the shutter wheel might need to be decreased somewhat from that used for the "instant acting" photo controls. If "slow acting" photo controls are to be tested, the rotational speed of the shutter wheel would best be controlled by a stepper motor. Based on a large data base, the average north-sky daylight increase and decrease can be expressed, in a slightly compressed form (one minute per equation equals two minutes actual), as a third-order polynomial function (see equations in **Section 4.6** and **Figure 4-1**).

To comply with actual vertical north-sky daylight conditions, about 18 minutes would be required to bring the light level from approximately 5 lux (0.5 fc) to about 120 lux (11 fc). By compressing the time constant to one-half of actual daylight characteristics, one full calibration procedure (one "on" and "off' cycle) can probably be performed in less than 25 minutes. A stepper-motor based design, using three different programs for the three types of photo controls, would simplify testing of all three types. (As an alternative to a stepper-motor controlled design for the "slow acting" photo controls, a much more complex shutter assembly was considered, but discarded after initial investigations.)

The 50-watt lamp employed in the present design, when used with the suggested LB-40 filter, provides sufficient light on the photoelectric control to check the calibration. When using the LB-80 filter to further increase the CCT, a higher wattage lamp, requiring a larger power supply, would be required. However, for the sake of uniformity, a Hoya LB-40 filter is recommended. Transmittance of the opal glass used was approximately 60% (measured with the glass held against a photometric head).

The drawings and schematics in this Annex have been used to successfully construct a test box, an individual user may design and build his or her own apparatus. However, for the results between different laboratories and purchasing agencies or specifiers to be reasonably alike, the basic design concept, as shown in **Figures A-1** through **A-5** (see **Section A.3**), should be followed.

### A.3 Drawings

The five figures in this section consist of four drawings for a light sensor calibration box (test box), plus a schematic for the illuminance meter interface. **Figures A-1**, **A-2**, and **A-3** show the details of the individual parts used in constructing the test box; **Figure A-4** is an assembly drawing. **Figure A-5** shows the schematic used to interface the illuminance meter with the photoelectric control device under test. This interface was designed and built in-house. It was used to replace the "hold button" of the Minolta T-1 illuminance meter used in this setup. Individuals versatile in electronics may develop their own designs. This circuit served to lock the illuminance value read by the illuminance meter display when the photoelectric control relay closes until the next event (relay opens) takes place. This gives the operator time to record the turn-on and turn-off settings, which can also be done automatically by downloading this information into a recording device. **Figure 4-1** (in **Section 4.6**) illustrates the increase of daylight based on a large data set and its mathematical expression.

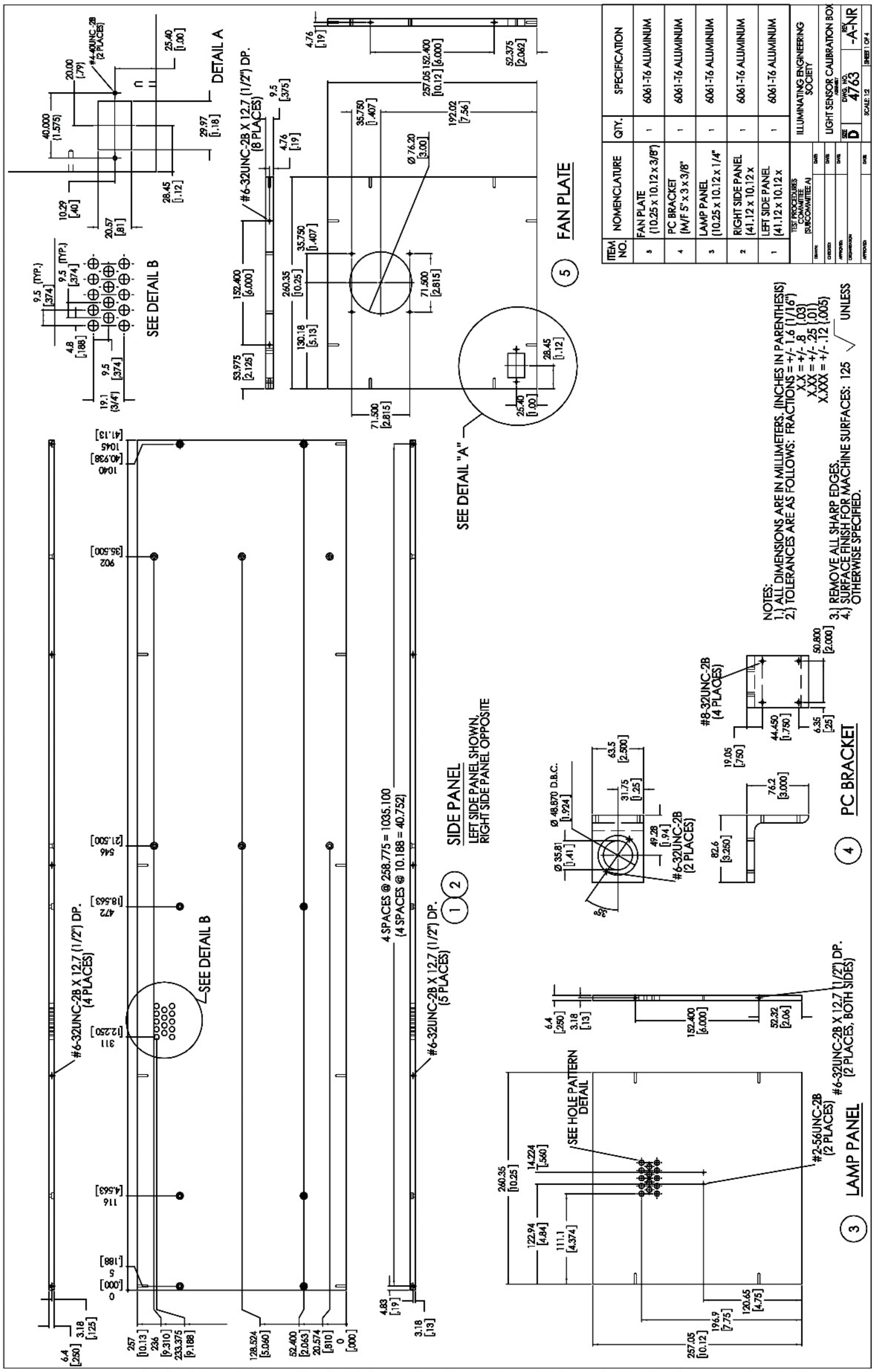



**Figure A-1: Mechanical specifications for the side panel, lamp panel, fan plate, and PC bracket that are part of the test box, with an associated materials list.**

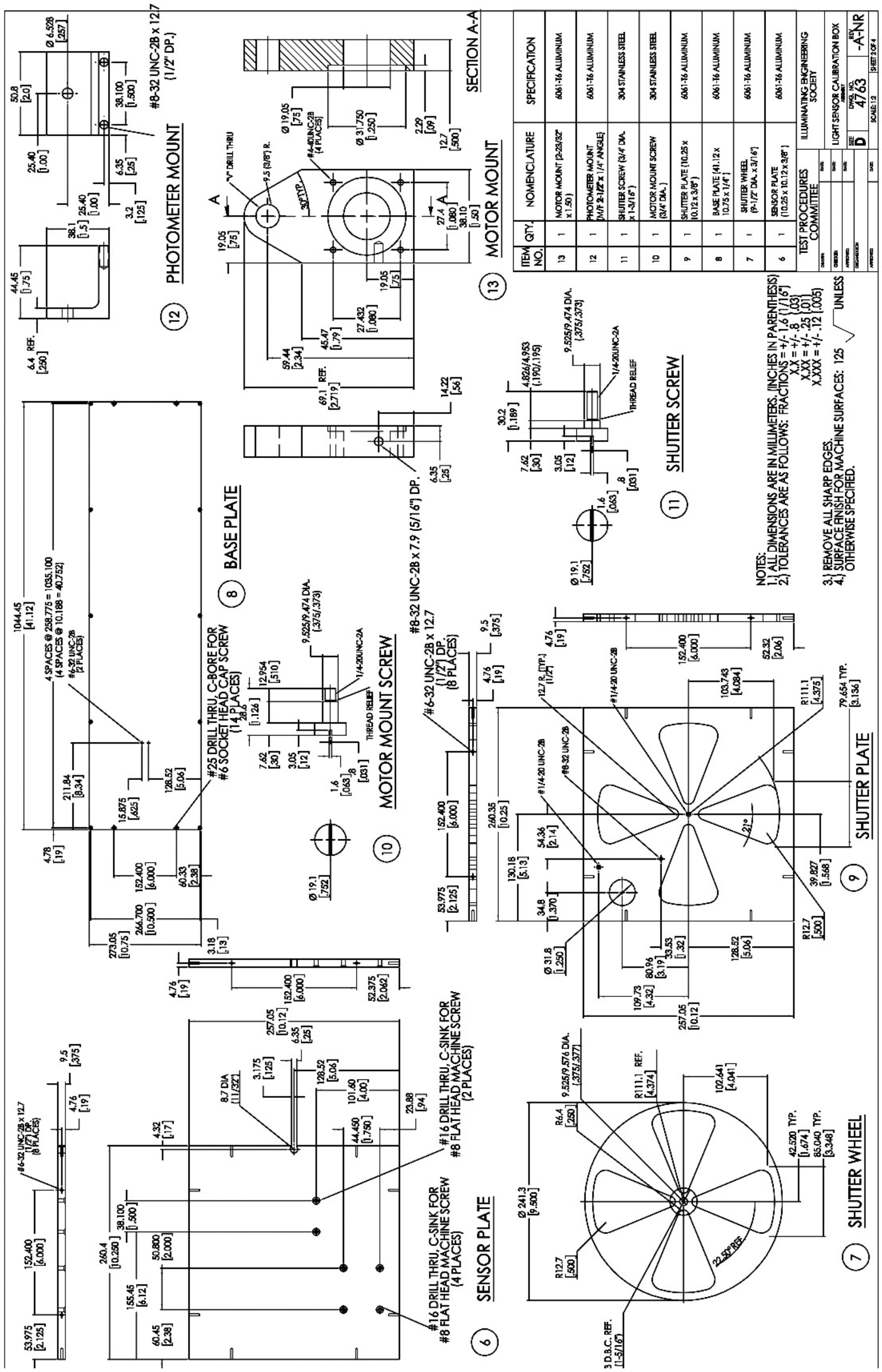



**Figure A-2. Mechanical specifications for the sensor plate, shutter wheel, base plate, shutter plate, motor mount screw, shutter screw, photometer mount, and motor mount that are part of the test box, with an associated materials list.**

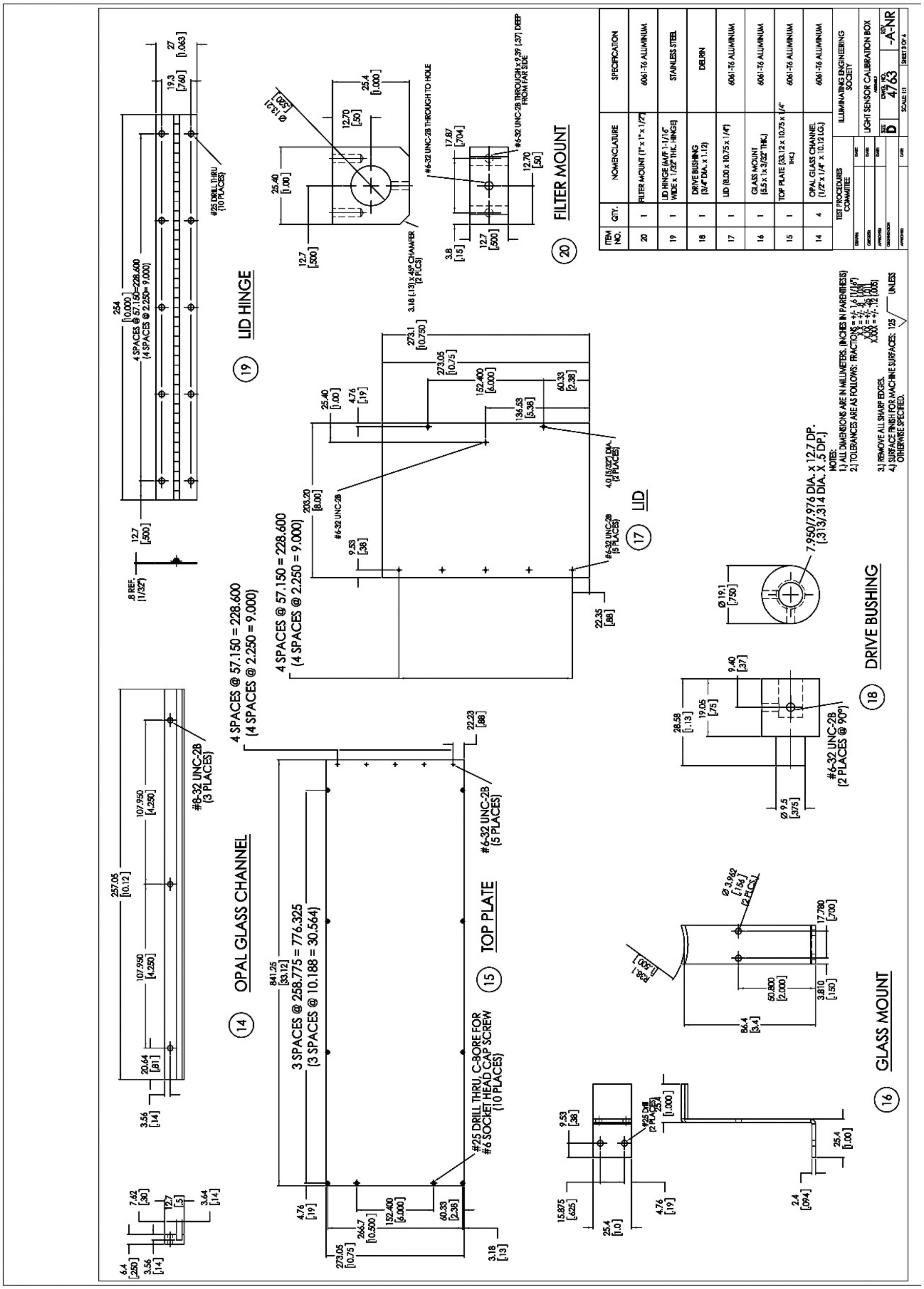
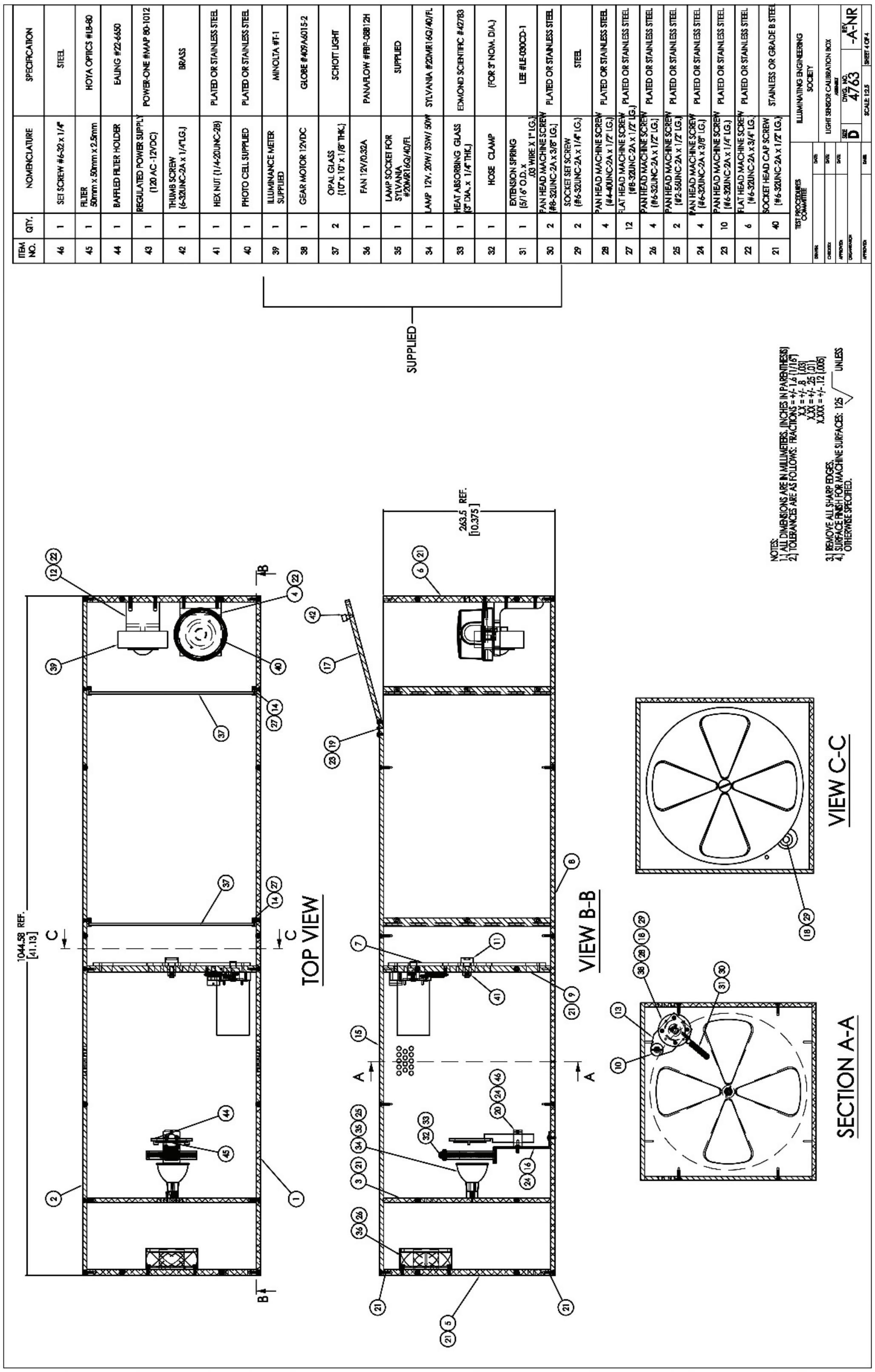



**Figure A-3. Mechanical specifications for the optical glass channel, top plate, glass mount, lid, drive bushing, and lid hinge that are part of the test box, with an associated materials list.**




| ITEM NO. | QTY. | NOMENCLATURE | SPECIFICATION |
|---|---|---|---|
| 46 | 1 | SET SCREW #6-32 x 1/4" | STEEL |
| 45 | 1 | FILTER 50mm x 50mm x 2.5mm | HOYA OPTICS #LB-80 |
| 44 | 1 | BAFFLED FILTER HOLDER | EALING #22-6650 |
| 43 | 1 | REGULATED POWER SUPPLY (120 AC - 12VDC) | POWER-ONE #MAP 80-1012 |
| 42 | 1 | THUMB SCREW (6-32UNC-2A x 1/4"LG.) | BRASS |
| 41 | 1 | HEX NUT (1/4-20UNC-2B) | PLATED OR STAINLESS STEEL |
| 40 | 1 | PHOTO CELL SUPPLIED | PLATED OR STAINLESS STEEL |
| 39 | 1 | ILLUMINANCE METER SUPPLIED | MINOLTA #T-1 |
| 38 | 1 | GEAR MOTOR 12VDC | GLOBE #409A6015-2 |
| 37 | 2 | OPAL GLASS (10" x 10" x 1/8" THK.) | SCHOTT LIGHT |
| 36 | 1 | FAN 12V/0.32A | PANAFLOW #FBP-08B12H |
| 35 | 1 | LAMP SOCKET FOR SYLVANIA #20MR16Q/40/FL | SUPPLIED |
| 34 | 1 | LAMP 12V, 20W/35W/50W | SYLVANIA #20MR16Q/40/FL |
| 33 | 1 | HEAT ABSORBING GLASS (3" DIA. x 1/4" THK.) | EDMOND SCIENTIFIC #42783 |
| 32 | 1 | HOSE CLAMP | (FOR 3" NOM. DIA.) |
| 31 | 1 | EXTENSION SPRING (5/16" O.D. x .03 WIRE X 1" LG.) | LEE #LE-030CD-1 |
| 30 | 2 | PAN HEAD MACHINE SCREW (#8-32UNC-2A x 3/8" LG.) | PLATED OR STAINLESS STEEL |
| 29 | 2 | SOCKET SET SCREW (#6-32UNC-2A x 1/4" LG.) | STEEL |
| 28 | 4 | PAN HEAD MACHINE SCREW (#4-40UNC-2A x 1/2" LG.) | PLATED OR STAINLESS STEEL |
| 27 | 12 | FLAT HEAD MACHINE SCREW (#8-32UNC-2A x 1/2" LG.) | PLATED OR STAINLESS STEEL |
| 26 | 4 | PAN HEAD MACHINE SCREW (#6-32UNC-2A x 1/2" LG.) | PLATED OR STAINLESS STEEL |
| 25 | 2 | PAN HEAD MACHINE SCREW (#2-56UNC-2A x 1/2" LG.) | PLATED OR STAINLESS STEEL |
| 24 | 4 | PAN HEAD MACHINE SCREW (#6-32UNC-2A x 3/8" LG.) | PLATED OR STAINLESS STEEL |
| 23 | 10 | PAN HEAD MACHINE SCREW (#6-32UNC-2A x 1/4" LG.) | PLATED OR STAINLESS STEEL |
| 22 | 6 | FLAT HEAD MACHINE SCREW (#6-32UNC-2A x 3/4" LG.) | PLATED OR STAINLESS STEEL |
| 21 | 40 | SOCKET HEAD CAP SCREW (#6-32UNC-2A x 1/2" LG.) | STAINLESS OR GRADE B STEEL |

**Figure A-4. Assembly drawing of the test box, with an associated materials list.**

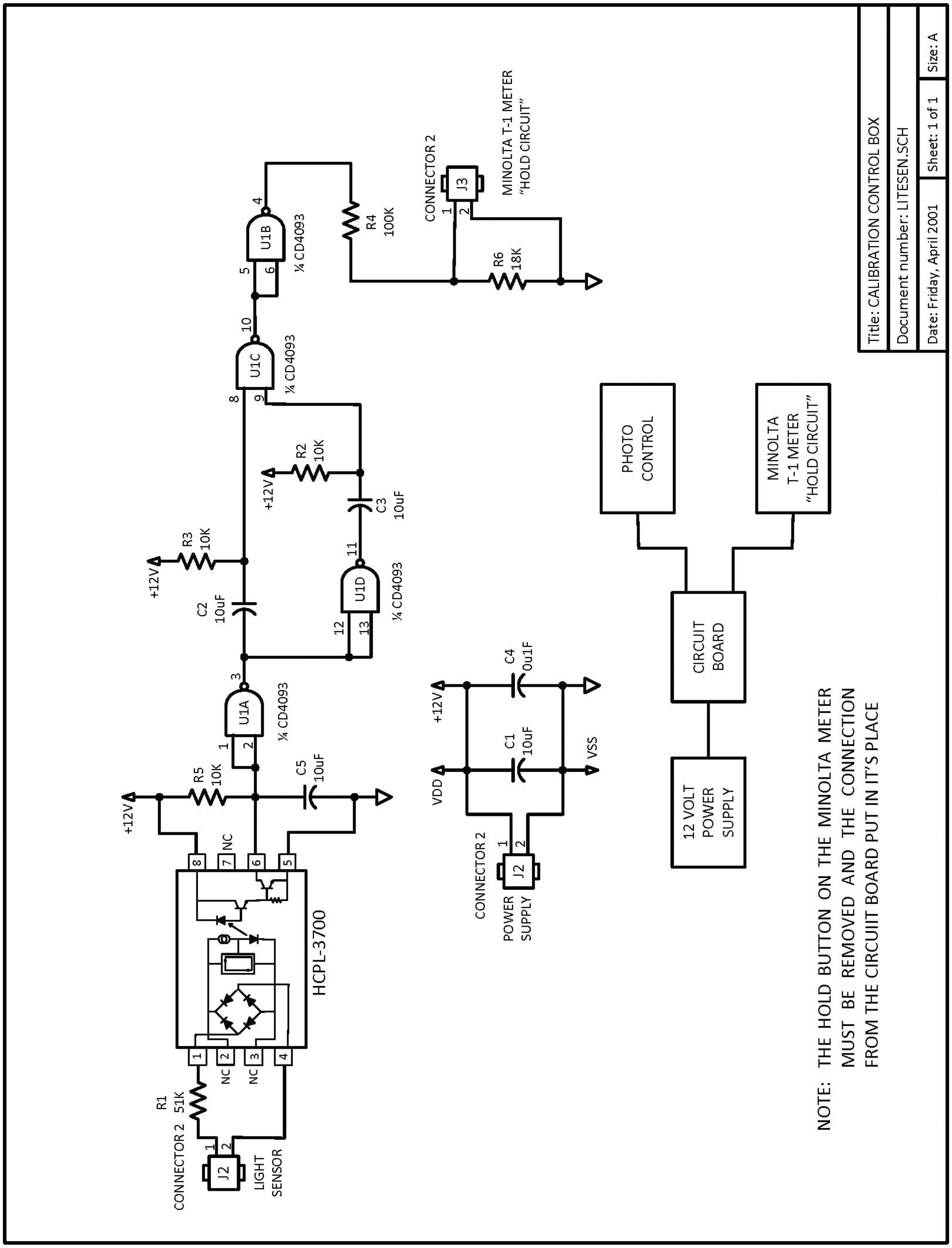



**Figure A-5. Schematic for interfacing the illuminance meter with the photoelectric control device under test.**

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